Side A: the row
`ic_online_closure_indexlift_r770`
source 1d328b2e… (PR #997's frozen digest, re-derived and checked)
state loop: one field inversion per stored state
insert loop: canonicalize root i, insert root i
self-test: full Feistel bijection check, 2 keys

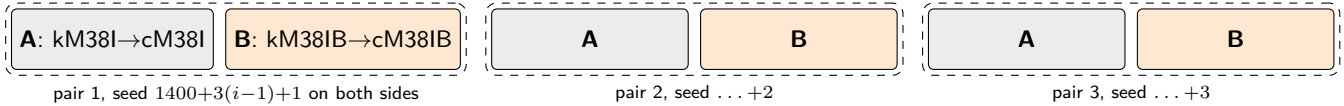
→

Side B: derived `ic_online_indexbuild_r770`
source bc6a31c7… = 1d328b2e… + 7 counted edits (invertible)
state loop: `RootBatch128`, one inversion per (ℓ, r) row
insert loop: canonicalize root $i{+}32$, prefetch its slot, insert root i
self-test: the same full check, 8 images at a time

Same output, gated per process and per base (K600, M38, M10): stored_states_blake3, table_slots_blake3, logical-state and table-map digests, root entries; the same relation, probes and scalar per point; the same rank tables; per rank seed the same relation digest, probe vector, root calls and counters (kM38I, kM38IB, kM38IBaa).

Segment Pi (point s$(i{-}1)$ of 12, role indexbuild-cold-v1); sub-block order shuffled by `random.Random(2026102050+`$i{-}1$`)`

primary: 3 pairs, strict alternation ABABAB or BABABA (seeded start; 7 points start A, 5 start B)



A/A: 2 pairs, alternating, seeds of primary pairs 1, 2

planted: 1 cold pair (lift_plant)

rho



Run order: O1 (Q0 on cPI, cPIB, cM10I, cM10IB, cM38I, cM38IB: both producers on all three bases; exactness only) then P1 … P12; 282 fresh processes, each pinned to one CPU of one ubuntu-24.04 runner; time guard at 165 min, job limit 180 min (any skip: invalid). Side = rank process (rank stage with the pair's seed, then the point) followed by cold process (verified table, then the point). Same rank seed on both sides of a pair, not a shared rank stage.

Primary (frozen): $E2E(\text{side}) = \text{rank_stage}(k) + \text{cold}(c)$, $R = \dfrac{\sum_{36\ \text{pairs}} E2E(A)}{\sum_{36\ \text{pairs}} E2E(B)}$; 95% two-stage paired bootstrap (points, then pairs within points; A and B together), 10,000 draws, seed 20261030. **Improvement iff the lower bound > 1.** Stage secondary (labelled): index build cM38I / cM38IB. Controls: A/A $E2E(B)/E2E(Baa)$ and cold inside $[0.97, 1.03]$; planted index build cM38IBX/cM38IB, lower bound > 1; planted closure corruptions rejected; planted non-bijections rejected (crate tests). Every gate true (unknown = fail).